IRAN PROSPERITY PROJECT — STRATEGIC REFERENCE DOCUMENT

Iran National Digital Identity System

سیستم هویت دیجیتال ملی ایران

PRODUCT REQUIREMENTS & SYSTEM DESIGN DOCUMENT

A sovereign, privacy-preserving digital identity infrastructure for post-transition Iran — covering 88 million domestic citizens and 8–10 million diaspora across 50+ countries. Built on W3C DID, Verifiable Credentials, and zero-knowledge proofs.

VERSION

2.0 — Definitive Edition

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PART I: EXECUTIVE SUMMARY

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1. Executive Summary

INDIS — the Iran National Digital Identity System — is a complete, sovereign, open-source digital identity infrastructure designed specifically for the requirements of post-transition Iran. It answers the single most critical operational question of the transition period: **how does a new government know who its citizens are, and how do citizens prove their identity to access services and participate in democratic processes — without recreating the surveillance apparatus of the regime they just removed?**

The answer is a system built on mathematical guarantees rather than institutional trust. INDIS uses zero-knowledge cryptography — the same technology used in international financial privacy systems — so that a citizen can prove they are over 18, or eligible to vote, or entitled to a pension payment, without revealing any other information about themselves to anyone. The government cannot track which shops a citizen visits. The pension office cannot see whether someone voted. The electoral commission cannot see a voter's home address. These are not policy promises — they are mathematical properties of the system's design.

INDIS is not a concept document. The core system has been built. All 15 backend services, the zero-knowledge proof engine, four blockchain chaincodes, a government operations portal, native mobile applications for Android, iOS, and HarmonyOS, a progressive web app, a diaspora enrollment portal, and a USSD gateway for feature-phone access are complete. The system is ready for infrastructure deployment, security audit, and staged rollout beginning on Day 1 of the transition.

The system covers +90 million domestic citizens and 8–10 million diaspora across 50+ countries. It includes enrollment pathways for every segment of Iranian society — including the estimated 2–3 million undocumented citizens in border regions who were systematically excluded from civil registration by the Islamic Republic, for whom social attestation (community co-signing) provides a legally valid path to identity.

What INDIS delivers to the transitional government:

Capability	Timeline	Strategic Value
Personnel vetting system	Day 40	Security sector verification; removes regime infiltrators
Pension ghost elimination	Day 90	Recovers an estimated 5–8% of pension budget from fraudulent beneficiaries
Referendum-ready voter rolls	Month 4	Cryptographically verifiable; internationally auditable
Anonymous testimony platform	Day 60	Enables Truth Commission without witness intimidation risk
		Replaces 40+ fragmented ministry identity systems

Capability	Timeline	Strategic Value
National service delivery layer	Month 12	
Full population coverage	Month 24	Universal digital identity across all 31 provinces

This document is the authoritative reference for the system — covering what it does, how it works, who uses it, what decisions are required from leadership, and how it deploys across the transition timeline.

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2. The Problem INDIS Solves

2.1 The Identity Crisis at the Heart of Transition

Every transitional government faces the same foundational challenge: it inherits the identity infrastructure of the regime it replaces. In Iran's case, that infrastructure has three critical defects that make it unsuitable for a democratic transition.

First, it was designed for control, not service. The Islamic Republic built identity systems to track, monitor, and restrict citizens — not to serve them. Every database was a surveillance instrument. Every credential check was a loyalty assessment. Continuing to operate these systems after transition means inheriting not just the databases, but the architecture of repression embedded in them.

Second, it is fragmented and untrustworthy. There is no single, authoritative source of Iranian identity. Civil registry data is incomplete and inconsistently maintained. Pension rolls contain an estimated 15–20% ghost beneficiaries who have been dead for years. Voter rolls have never been independently verified. The Ministry of Interior, Ministry of Health, Ministry of Finance, and Ministry of Justice each hold different, conflicting records for the same citizens.

Third, it excluded millions of people. Systematic exclusion from civil registration affected ethnic and religious minorities, rural border communities, and children of undocumented parents. These citizens exist — they have lived in Iran for generations — but the state denied their existence because it was convenient. A legitimate transitional government cannot simply inherit this exclusion.

2.2 The Surveillance Dilemma

There is a second, equally serious problem: the risk of building a new surveillance state in the name of building a democratic one.

Any centralized identity database — even one run by a well-intentioned government — is a potential future instrument of oppression. The next government, or the one after that, can always find a pretext to misuse it. History is full of identity systems built by democratic reformers that were later used by their successors for persecution.

INDIS resolves this dilemma not through policy or legal restriction (which future governments can repeal), but through cryptographic architecture. When a citizen uses INDIS to verify their voter eligibility, the cryptographic system generates a mathematical proof that they are eligible — and transmits only that proof. No location data. No name. No voting history. No pattern of service usage. These facts are not merely "not recorded" — the system is architecturally incapable of recording them, because they are never transmitted.

This is the core innovation of INDIS and the reason it merits serious consideration by any government that is serious about building a genuinely free society.

2.3 The International Legitimacy Problem

The transitional government will need to demonstrate to the world that:

- The referendum was free and fair
- Pension and subsidy payments went to real people
- Security clearances were issued through a verifiable process, not nepotism
- Transitional justice proceedings were conducted with authenticated evidence

Without a trustworthy identity infrastructure, none of these claims can be substantiated. With INDIS, every one of them is cryptographically provable and independently auditable by international observers.

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3. Why This Matters for the Transition

The Emergency Phase Booklet identifies five domains where identity infrastructure is an immediate operational requirement. INDIS directly addresses all five.

3.1 Political Chapter — Referendum Within 4 Months

The referendum is the political legitimacy event of the transition. Its credibility depends entirely on the integrity of the voter roll and the verifiability of the voting process.

INDIS provides:

- **A voter roll that can be independently verified by any international observer** — not because they trust the government's word, but because the cryptographic audit trail is publicly readable
- **ZK-STARK electoral proofs** — the strongest available cryptographic tool for election verification, with post-quantum security guarantees

- **Anti-coercion architecture** — the receipt-free voting system is designed so that a citizen cannot prove to a third party how they voted, eliminating vote-buying and workplace coercion
- **Diaspora participation** — the embassy enrollment network enables Iranian citizens in 50+ countries to participate in the referendum through a verified, tamper-evident process
- **USSD fallback** — citizens in rural areas with no smartphone can check their voter eligibility via feature phone, using a national short code

The electoral module meets the operational requirement: fully functional and independently audited by international firms at least 14 days before the referendum date.

3.2 Macroeconomic Chapter — Fiscal Stabilization

The pension and subsidy system is a critical early test of the transitional government's competence and good faith. Ghost beneficiaries — deceased individuals whose pension payments continue to flow to family members or fraudulent actors — represent an estimated 5–8% of the pension budget. At the scale of Iran's pension system, this is a multi-billion dollar annual loss.

INDIS eliminates this through biometric deduplication. When a citizen enrolls, their fingerprint and facial geometry are checked against every other enrolled citizen. The same person cannot appear twice. After enrollment, pension payment is conditional on credential validity — a dead citizen's credential expires and cannot be renewed by anyone else, because no one else shares their biometric signature.

Within 90 days of a full enrollment campaign for pension-age citizens, the ghost beneficiary rate drops to near zero. The fiscal impact is immediate and permanent.

3.3 Military Chapter — Personnel Vetting by Day 40

The security sector is the highest-stakes environment for the transition. Regime remnants embedded in the military and security services are a direct threat. Identifying and vetting personnel before they have the opportunity to organize or cause harm requires a rapid, trustworthy credentialing system.

INDIS provides a Security Clearance credential that can be enrolled in bulk for military and civil service personnel. The credential is biometrically bound — it cannot be transferred or forged. The government portal gives authorized officers the ability to verify clearance status in real time at any checkpoint.

The Day 40 deadline is achievable: the bulk enrollment pipeline can process 500,000 credentials per day, covering the entire priority population in the first two weeks.

3.4 Transitional Justice — Truth Commission by Day 60

The Truth Commission faces a design challenge that has no good analog in other transitional systems: witnesses need to testify without facing retaliation from regime remnants who may still have operational capability, while the Commission needs to verify that testimony is coming from genuine Iranian citizens rather than planted disinformation.

INDIS provides both simultaneously, through a zero-knowledge anonymous testimony system:

- A witness submits testimony and proves cryptographically that they are an Iranian citizen — without revealing their name, location, or any other identifying information
- The Commission receives a receipt token that the witness can use to link follow-up submissions to the original, without ever de-anonymizing themselves
- If the witness chooses to reveal their identity to a sealed court record — and only then, with explicit written consent — the system facilitates that disclosure under judicial escrow

This architecture makes intimidation structurally impossible. There is no list of witnesses to target. The system does not know who they are.

3.5 Government Essential Functions — Service Continuity

The transitional government will be expected to deliver services from Day 1: pensions, healthcare coverage verification, education enrollment, banking access, and dozens of other functions. Each of these currently runs on a separate, fragmented system with no interoperability and no common identity standard.

INDIS provides the unifying layer: a single credential wallet that citizens can use to access every government service, with role-based access controls that prevent any ministry from accessing data outside its authorized scope.

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4. Value Delivered by Layer

4.1 Value to Citizens

Benefit	Current State	With INDIS
Proving identity	Long queues; physical documents; multiple visits per service	Single QR scan; proof generated on their phone in <5 seconds
Privacy	Government tracks every service interaction	Verifiers see only a boolean result; government has no behavioral profile
Inclusion	2–3 million undocumented citizens locked out of services	Social attestation pathway provides a valid credential for every Iranian
Rural access	Requires travel to provincial offices	USSD short code; mobile enrollment units; offline capability
Diaspora	Embassy bureaucracy; no digital path	Full digital enrollment through embassy network; same credential parity

Benefit	Current State	With INDIS
Trust in government claims	No way to verify government statistics	Every claim (pension roll, voter count) is cryptographically provable

4.2 Value to Government

Function	Current Cost	With INDIS
Pension ghost elimination	~5–8% of budget lost	Biometric deduplication removes duplicates at enrollment; savings are permanent
Ministry identity integration	40+ separate systems; no interoperability	Single identity layer; ministries access what they need, nothing more
Election administration	Manual voter roll verification; disputed results	Cryptographic voter rolls; independently auditable; internationally recognized
Security sector vetting	Manual background checks; unreliable records	Biometrically bound credentials; tamper-evident; bulk-issued in 48 hours
International credibility	Identity system inherited from regime	W3C standards-compliant; internationally auditable; open-source and verifiable

4.3 Value to International Partners

Partner Need	INDIS Capability
Referendum verification	ZK-STARK electoral proofs; public audit trail; observer API access
AML/KYC for sanctions relief	Verified citizen credentials for banking re-integration
Humanitarian aid targeting	Verified beneficiary credentials without surveillance exposure
Diplomatic recognition readiness	ICAO 9303-compliant physical identity cards recognized by 197 countries
Development finance	Verifiable population data for project scoping and beneficiary targeting

5. What Makes INDIS Different

The identity systems used by other governments fall broadly into two categories: centralized databases (which concentrate power and invite abuse) and fragmented systems (which fail at scale and produce inconsistent results). INDIS is neither.

5.1 Compared to Legacy Government ID Systems

Most government identity systems work like this: a citizen presents documents, the government stores their data in a database, and when verification is needed, the database is queried. This means:

- The government has a complete behavioral profile of every citizen
- A database breach exposes all citizen data
- Foreign governments or companies with database access can conduct mass surveillance
- The system is a single point of failure

INDIS works differently. The government stores only cryptographic hashes and public keys — never raw identity data on any server that can be queried remotely. Credentials live on the citizen's device. Verification is a mathematical computation that produces a yes-or-no answer without data exchange.

5.2 Compared to UAE Pass and Similar Regional Systems

UAE Pass is the closest regional analog. It is a well-implemented centralized identity system. INDIS exceeds it on three dimensions:

Dimension	UAE Pass	INDIS
Privacy Model	Centralized data store; government has full behavioral data	ZK-proof selective disclosure; no behavioral profile possible
Verification	Reveals identity attributes to verifiers	Reveals only a boolean result; zero personal data
Inclusivity	Urban smartphone-first	Offline-capable, multilingual, social attestation, USSD/feature phone
Post-Quantum Security	Not addressed	CRYSTALS-Dilithium3 for long-term credentials
Sovereignty Architecture	UAE government has administrative access	No single entity has administrative access; multi-party key ceremony
Auditability	Internal audit	Open-source; public audit trail; international observer access

5.3 The Privacy Architecture Is Not Optional

Some government advisors will suggest that a surveillance-capable architecture is desirable for security purposes — that the government should be able to see citizen behavior to detect threats.

This argument should be rejected on both principled and practical grounds.

Principally: A government that builds surveillance infrastructure is not building a free society. It is building the next phase of the same system the transition was supposed to end.

Practically: Surveillance capability is a liability, not an asset. It makes the system a high-value target for foreign intelligence agencies. It exposes the government to international criticism. It creates the political conditions for the next authoritarian to claim a ready-made population control infrastructure.

The privacy architecture of INDIS is a feature, not a limitation. It is what makes the system trustworthy to citizens, credible to international partners, and safe for long-term democratic governance.

PART II: STAKEHOLDER MAP

6. Stakeholder Map

INDIS serves six distinct stakeholder categories, each with different needs, different access levels, and different relationships to the system's trust model.

6.1 Citizens — +90 Million Residents

Citizens are the primary users and the primary beneficiaries of INDIS. They interact with the system through the mobile app (Android, iOS, HarmonyOS), the Citizen PWA (browser-based), USSD short codes (feature phones), kiosk terminals (post offices, hospitals), and physical identity cards.

Citizen segments:

Segment	Size (est.)	Primary Channel	Key Requirement
Urban smartphone users	~50M	Mobile app / PWA	Fast, private, full-featured
Rural / low-connectivity	~20M	USSD, mobile enrollment units, kiosk	Offline-capable; works on feature phones
Elderly	~10M	Assisted enrollment, simplified mode, physical card	Large text; audio guidance; physical backup
Youth and students	~15M	Mobile app	Transparent privacy controls; referendum participation
	~2–3M		

Segment	Size (est.)	Primary Channel	Key Requirement
Undocumented / documentation-deficient		Social attestation via mobile enrollment units	Non-exclusionary; basic services immediately
Iranian diaspora	~8–10M	Embassy network; diaspora portal	Remote enrollment; full credential parity; referendum access

The system's fundamental design principle regarding citizens: **citizens own their identity**. The government is the issuer of credentials, not the custodian of data. A citizen's private key is generated on their device and never transmitted to any server under any circumstances.

6.2 Government Ministries and Authorities

Government entities interact with INDIS through the Government Portal — a secure, certificate-authenticated web application that provides ministry-specific dashboards, credential issuance workflows, audit access, and bulk operations.

Ministry / Authority	Primary Use	Data Access Scope
National Identity Authority (NIA)	System administration; credential issuance	Administrative (limited by multi-party controls)
Ministry of Interior	Civil registration; elections	Citizenship, Residency, Voter Eligibility
Ministry of Finance	Pension; tax identity	Pension, Citizenship
Ministry of Health	Coverage verification; prescriptions	Health Insurance
Ministry of Education	Student enrollment; teacher credentialing	Age Range, Professional
Ministry of Justice	Court records; proceedings	Full Identity — Level 3, judicial authorization required
Ministry of Foreign Affairs	Passport management; diaspora	Citizenship, Diaspora
Electoral Commission	Voter roll management; referendum	Voter Eligibility
Transitional Divan (court)	Proceedings authentication	Amnesty Applicant, Full Identity
Transitional Mehestan	Member credentialing; oversight	Voter Eligibility, Security Clearance

Ministry / Authority	Primary Use	Data Access Scope
NISS / Military Command	Personnel vetting; access control	Security Clearance
Central Bank	AML compliance; beneficial ownership	Citizenship, Full Identity Level 3

Critical access control principle: Role-based access is enforced at the system level, not through policy. A pension officer logging into the portal cannot see security clearance data. A healthcare administrator cannot see electoral data. These restrictions are not overridable by any individual ministry official — they require multi-party authorization and leave an immutable audit trail.

6.3 Verifiers — Organizations Checking Credentials

Verifiers are organizations — public or private — that need to confirm aspects of a citizen's identity to deliver a service. Critically, most verifiers should receive less information than they currently get, not more.

Verification tiers:

Level	Method	What Verifier Receives	Example Use
Level 1	QR scan + ZK proof	Boolean: eligible / not eligible	Age check at pharmacy; voter eligibility at polling station
Level 2	NFC chip + biometric match	Credential category + validity date	Banking KYC; border crossing
Level 3	Full identity disclosure	Complete verified identity — with explicit citizen consent captured	Court proceedings; emergency medical care
Level 4	Emergency override	Full identity; no citizen consent required	Active security emergency; requires senior multi-party authorization + automatic audit alert to Parliamentary Committee

Private sector verifier program: Banks, insurance companies, employers, telecommunications providers, and real estate registries can register as authorized verifiers. Each verifier receives a certificate defining exactly which credential types it may request and at what frequency. Verifier authorization is public record.

6.4 Enrollment Agents

Enrollment agents are trained personnel who operate the enrollment hardware — biometric capture devices, document scanners, and the enrollment agent application. They are the human interface between citizens (particularly those who cannot self-enroll) and the system.

Agent Type	Location	Special Capabilities
Fixed center agents	Provincial offices, post offices, hospitals, universities	Full enrollment including biometric capture
Mobile unit agents	Rural areas, border regions, displaced communities	Same as fixed, satellite-connected, generator-powered
Embassy agents	50+ countries	Diaspora enrollment with embassy certificate authority
Community agents	Remote villages	Initiate social attestation; schedule biometric mobile unit visit

All agent actions are logged with agent's credential and timestamped. Supervisory review is required for social attestation enrollments. Anomaly detection flags unusual enrollment patterns for human review.

6.5 International Partners and Observers

International observers — whether from multilateral organizations, allied governments, or civil society — need to verify the integrity of the system without having access to citizen data.

INDIS provides a dedicated observer API that allows:

- Public audit of the blockchain anchor log (hash-only, no personal data)
- Aggregate statistical reporting with differential privacy guarantees
- Electoral result verification via publicly available ZK verification keys
- Real-time system health monitoring without data access

The observer interface is designed specifically to satisfy the verification requirements of international election monitoring bodies (ODIHR, Carter Center, UN Electoral Assistance Division) and financial intelligence organizations (FATF, Wolfsberg Group).

6.6 System Administrators and Auditors

Role	Access	Safeguards
NIA System Administrators	System configuration and key ceremony	Multi-party authorization for any sensitive operation; separation of duties enforced
Independent Auditors	Read-only system logs and cryptographic proofs	No access to citizen data; observer API only
Parliamentary Oversight Committee	Aggregate statistics; audit reports	No individual citizen data; monthly public reporting required
Security Red Team	Controlled penetration testing environment	Scheduled, isolated; findings reported to Parliamentary Committee

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7. User Journeys

7.1 The Urban Citizen — Self-Enrollment in 15 Minutes

Maryam lives in Tehran. She has a smartphone and a national ID card. She has heard about INDIS on state television and wants to enroll.

Step 1 — Download and Setup (2 minutes)

She downloads the INDIS app from the official app store. The app opens in Persian. She reads the privacy notice — written in plain language, not legal boilerplate — which explains exactly what data will be captured, where it goes, and what rights she has. She gives explicit consent by signing with her fingerprint.

Step 2 — Document Capture (3 minutes)

She photographs her national ID card. The app's AI checks that the photo is clear and authentic. She can also photograph a passport or driver's license — any combination of official documents is accepted. If a document is not legible, the app guides her through retaking the photo.

Step 3 — Biometric Capture (5 minutes)

The app guides her through a facial recognition sequence with liveness detection — she follows simple on-screen prompts (look left, look right, blink) to prove she is physically present. Then she photographs each finger. The process is audio-guided so she does not need to read instructions.

Step 4 — Processing (3 minutes)

The app submits her enrollment. In the background, the system runs biometric deduplication against the national database. Her DID — a unique cryptographic identifier — is generated on her phone. The private key never leaves her device.

Step 5 — Credential Issuance (2 minutes)

The government issues her verifiable credentials: Citizenship, Age Range (31–50, without revealing her exact age), Residency (Tehran, urban), Health Insurance status. Her digital identity card appears in the wallet screen. A notification is sent confirming enrollment.

She is now enrolled. If she ever loses her phone, she can recover her credentials through a secure recovery process using biometric re-verification at an enrollment center. Her private key is backed up in encrypted form to a key she holds — not to a government server.

7.2 The Rural Undocumented Citizen — Social Attestation

Ali is a 45-year-old farmer in a Kurdish border village in Kurdistan Province. He has no documentation. His parents were never registered. He was born at home, never registered at a hospital or civil registry office. By the Islamic Republic's reckoning, he does not exist.

A mobile enrollment unit arrives in his village. The unit is staffed by two agents who speak Kurdish and Persian, and carries its own satellite internet connection and power supply.

Step 1 — Community Attestation

Three members of his village who are already enrolled in INDIS — his neighbor, the village elder, and a local teacher — co-attest his identity. They each use the app to sign a digital statement: "I personally know this person. His name is Ali [family name]. He has lived in this village for at least 20 years." Each signature is cryptographically bound to the attestor's own verified credential.

Step 2 — Basic Information

The enrollment agent records what is known: his given name, approximate age (the system accepts "approximately 40–50" if the exact birth year is unknown), village of origin, and a description of family relationships where known.

Step 3 — Biometric Capture

The agent captures his photograph and fingerprints on the mobile unit's hardware. This biometric record is now his cryptographic anchor — even without documents, no one can impersonate him.

Step 4 — Social Attestation Credential Issued

He receives a Social Attestation credential. It is marked as Tier 2 (below Standard and Enhanced enrollment), which means it grants access to healthcare and basic services immediately, but voting eligibility requires a secondary review process. The app shows him the upgrade path — what documentation or additional attestation would upgrade his credential to Standard.

He is now in the system. He exists. He can receive healthcare. When the pension system is connected, he will receive any benefits he qualifies for. For the referendum, his voting eligibility is subject to review — but that review process is defined, transparent, and has a clear appeal path.

The political importance of this moment cannot be overstated: the new government has just recognized a citizen whom the old government spent 45 years pretending did not exist.

7.3 The Diaspora Citizen — Remote Enrollment

Shirin lives in Paris and left Iran fifteen years ago. She holds a valid Iranian passport, expired by two years. She has strong opinions about the referendum and wants to participate.

She visits the INDIS diaspora portal from her laptop. The portal supports Persian, English, and French. She begins an enrollment session and photographs her expired passport — expired documents are accepted with a flag that they require embassy officer confirmation.

She schedules an appointment at the Iranian embassy in Paris. At the appointment, an embassy agent verifies her documents, captures her biometrics using the embassy's enrollment hardware, and completes her enrollment. Her Diaspora credential is issued with full parity to domestic credentials — she can vote in the referendum.

The embassy appointment took 20 minutes. She receives her digital identity card in the app that afternoon.

7.4 The Verifier — Age Check Without Seeing Age

A pharmacist at a Tehran pharmacy needs to verify that a customer is over 18 before dispensing a restricted medication. Under the old system, the pharmacist would ask for an ID card, look at the birth date, do the math, and hand the card back — having now seen the customer's full name, national ID number, home address, and exact date of birth.

With INDIS:

The pharmacist's terminal displays a QR code. The customer opens their INDIS app, selects "Prove age over 18," and taps confirm. Their phone generates a zero-knowledge proof in under 3 seconds. They scan the QR code. The pharmacist's screen shows a green checkmark: **Eligible — Over 18** .

The pharmacist has not seen the customer's name. Has not seen their national ID number. Has not seen their age. Has not seen their address. Has not seen their birth date. The only fact transmitted is the one fact needed: the customer is over 18.

This is the core value proposition of INDIS stated concretely: privacy is not a policy you promise. It is a mathematical property of the transaction.

7.5 The Pensioner — Ghost Elimination

Hossein, a retired civil servant, died three years ago. His family continued to receive his pension payments by reporting him alive to the system, which had no way to verify.

When INDIS enrollment runs for the pension-age population, the system attempts to enroll Hossein. His family cannot enroll him — they do not have his fingerprints. His credential was never issued. After a 90-day grace period during which uncredentialed pension recipients are required to enroll or have their payments reviewed, his payments are suspended pending investigation.

The system does not need to accuse anyone of fraud. It simply requires that pension recipients be real, enrolled, biometrically verified people. Anyone who is not can seek reinstatement through a formal review process.

Nationwide, this process eliminates an estimated 15–20% of fraudulent pension beneficiaries within three months.

7.6 The Witness — Anonymous Testimony

Leila witnessed a massacre. She knows names. She knows locations. She knows dates. She is terrified of retaliation from the perpetrators, several of whom are believed to still be operating.

She uses the INDIS anonymous testimony system. She proves — through a zero-knowledge proof — that she is an Iranian citizen. She does not reveal her name, her location, or her identity. She submits her testimony.

The Truth Commission receives her testimony along with a cryptographic proof that it came from a genuine, verified Iranian citizen. The Commission cannot discover who she is. Even if subpoenaed, the system does not hold information that would allow identification.

She receives a receipt token — a random cryptographic string. If she wants to submit follow-up testimony that can be linked to her original statement (e.g., "in addition to what I said before..."), she includes the token. The Commission can verify that both submissions came from the same person, without knowing who that person is.

If, at some later point, she chooses to reveal her identity to a sealed court record — only then, after signing an explicit consent form, witnessed by a legal representative — the system facilitates a controlled disclosure to a judicial escrow. Even then, the information is available only to the specific judicial body, not to the Commission as a whole.

PART III: CORE CAPABILITIES

8. Core Capabilities Overview

INDIS is organized around eight core capabilities that together constitute a complete national identity infrastructure.

Capability	Delivered By	Status
Enrollment — getting every Iranian into the system	Enrollment service + mobile apps + kiosk + USSD	Complete
Credential wallet — the digital identity card	Identity + credential services + mobile apps	Complete
Verification — proving claims without revealing data	ZK proof engine + verifier app	Complete
Electoral module — referendum and election engine	Electoral service + STARK proof system	Complete
Transitional justice — anonymous testimony + amnesty	Justice service + Bulletproofs	Complete
Government portal — ministry operations	Govportal service + React frontend	Complete (~98%)
Offline access — USSD, feature phones, offline proofs	USSD service + offline-capable mobile app	Complete
Physical card — ICAO 9303 machine-readable ID	Card service	Complete (NFC encoding deferred)

9. Enrollment

9.1 Three Enrollment Pathways

No single enrollment pathway can reach all 88 million Iranians. INDIS provides three, each designed for a different segment of the population.

Pathway 1 — Standard Enrollment (est. 75–80% of population)

Requires: any combination of official documents (national ID card, passport, birth certificate, driver's license) + biometric capture.

Documents are verified by AI-assisted authenticity checking, then human review for flagged cases. The process takes 10–20 minutes and can be completed fully through the mobile app with no in-person visit required for most citizens.

Pathway 2 — Enhanced Enrollment (est. 15–20% of population)

For citizens whose documents can be cross-referenced with existing civil registry data. The system performs an automated reconciliation with the civil registry database (Ministry of Interior records, birth registration archives) and issues a higher-trust credential. This pathway is handled by enrollment agents at fixed centers.

Pathway 3 — Social Attestation (est. 2–5% of population)

For citizens with no documentation and no civil registry record. Three enrolled community members (minimum Tier 2 credential holders) co-sign a digital attestation. Biometric capture is required. The resulting credential is Tier 2 (lower privilege than Standard), but grants immediate access to healthcare and basic services, with a clear upgrade path.

Social attestation is the system's most politically significant feature. It is the mechanism by which the transitional government fulfills its commitment to recognize every Iranian — including those the previous government chose not to.

9.2 Bulk Enrollment

For priority populations (military, civil service, healthcare workers), the system supports a bulk enrollment pipeline:

- Batch import of existing personnel records
- Biometric capture sessions at designated locations
- Credential issuance within 24 hours of biometric capture
- Department-level progress tracking in the Government Portal
- Throughput: 500,000 credentials per day under normal load

9.3 Biometric Deduplication

Every enrollment triggers a check against the entire enrolled population. The system computes a similarity score between the new enrollment's biometric template and every existing template in the database.

This check serves two purposes:

1. **Preventing duplicate enrollment** — the same person cannot appear twice in the system
2. **Detecting fraud** — biometric similarity flags for human review

The deduplication system targets:

- False Match Rate (one person matches as another): ≤ 0.0001%
- False Non-Match Rate (same person not recognized): ≤ 0.1%
- Time to complete: 30 seconds typical; 90 seconds maximum at full 88M population scale

Biometric templates are stored encrypted with AES-256-GCM, keyed by HSM-managed keys. They are never transmitted. The system operates on similarity scores, not raw templates, when performing cross-population comparison. Original biometric data cannot be reconstructed from stored templates.

10. Credential Wallet

10.1 The Digital Identity Card

The primary screen of the INDIS app shows a digital identity card — a visual representation of the citizen's verified identity, styled in Iranian cultural heritage visual language. The card displays:

- The citizen's name in Persian (Nastaliq script) with English transliteration
- A photograph (captured at enrollment)
- Verified status indicators for each credential held
- A biometric-gated QR code (the QR only activates after fingerprint or facial recognition)
- Current validity status

Sensitive fields — national ID number, exact birth date, home address — are masked by default. The citizen can reveal them by tapping and authenticating biometrically. This design prevents shoulder-surfing in public settings.

10.2 Credential Types

The system issues eleven credential types, each designed to provide the minimum necessary information for its intended use:

Credential	What It Contains	What It Does NOT Contain
Citizenship	Iranian citizen status, date of registration	National ID number, birth date, address
Age Range	Age bracket (e.g., 31–50)	Exact birth date or year
Voter Eligibility	Eligible for [specific election]: yes/no	Voting district (below province), address
Residency	Province, urban/rural	Street address

Credential	What It Contains	What It Does NOT Contain
Professional	Credential category, qualification level	Employer, salary, specific institution
Health Insurance	Coverage type, enrollment status	Medical history, diagnoses
Pension	Beneficiary status, payment eligibility	Payment amount, account details
Security Clearance	Clearance level only	Clearance basis, associated investigations
Amnesty Applicant	Application status, case reference	Crime details, victim information
Diaspora	Diaspora status, country of residence	Passport number, foreign address
Social Attestation	Attestation tier, co-attestors (pseudonymous), upgrade path	Attestor identities (unless consent given)

10.3 Credential Lifecycle

Every credential has a defined lifecycle:

```
Issue → Active → [Renewal or Expiry] → Expired
                    ↓
              Revocation (immediate, on-chain, propagates to verifiers in ≤60 seconds)
```

Citizens receive expiry notifications at 30 days, 7 days, and 1 day before expiry. Renewal is a streamlined process — biometric re-verification only, no document re-submission for citizens with an unbroken credential chain.

Revocation is anchored to the blockchain. Any verifier — online or offline with a cached revocation list — can check revocation status without querying a central server. Revocation propagates within 60 seconds to all online verifier nodes, and is available to offline verifiers within the 72-hour cache refresh cycle.

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11. Verification — Proving Claims Without Revealing Data

11.1 The Zero-Knowledge Principle in Practice

The verification flow is the operational heart of the privacy architecture. Understanding it is essential for evaluating INDIS.

When a verifier — whether a pharmacist, a bank officer, a border guard, or a polling station worker — needs to verify a fact about a citizen, the flow is:

1. The verifier's terminal sends a **verification request** to the citizen's device. The request specifies: what type of credential is being requested, what predicate must be satisfied (e.g., "age > 18"), and a random nonce to prevent replay attacks. **No citizen data is sent in the request.**
2. The citizen's device **generates a zero-knowledge proof on the device itself**. The proof is a mathematical object that encodes the answer to the question (yes, this citizen satisfies the predicate) without encoding any of the underlying data. Generation takes 2–5 seconds on a mid-range 2022 Android phone.
3. The citizen sees an approval screen: "Pharmacy [Name] is requesting proof of age over 18. Approve?" They confirm with fingerprint or face recognition.
4. The device sends **the proof only** to the verifier's terminal. No name, no ID number, no birth date, no address — just the mathematical proof.
5. The verifier's terminal checks the proof against the public verification key (published by NIA) and the credential's revocation status (checked against the blockchain). This takes approximately 200 milliseconds.
6. The terminal displays: [DONE] **Eligible — Over 18** or [NO] **Not Confirmed**.

The verifier has learned exactly one thing: whether the citizen satisfies the predicate. Nothing else. This is not achieved through trust or policy — it is achieved through mathematics.

11.2 Offline Verification

Verification works without network connectivity, within limits:

- The citizen's device generates ZK proofs fully offline — the proof engine runs locally
- The verifier's terminal uses a cached revocation list (up to 72 hours old)
- Credential anchors are stored locally after initial download

The 72-hour offline window covers planned outages and connectivity gaps, while ensuring that revoked credentials become unverifiable within three days in the worst case. For high-stakes verification contexts (border control, financial transactions), online verification is required.

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12. Electoral Module — The Referendum Engine

12.1 Cryptographic Election Properties

The electoral module is the most security-critical component of INDIS. It must satisfy six properties simultaneously — properties that historically have been in tension with each other:

Property	What It Means	How INDIS Achieves It
Completeness	Every valid vote is counted	Cryptographic commitment scheme; no vote can be silently dropped
Soundness	No invalid vote is counted	ZK-STARK proof required for every ballot
Ballot Privacy	No one can determine how any individual voted	Cryptographic ballot encryption; linkage to voter not possible
Individual Verifiability	Every voter can verify their vote was counted	Each voter receives a cryptographic receipt they can check
Universal Verifiability	Any mathematician can verify the total result	Publicly auditable ZK proof of aggregate; open-source verification code
Receipt-Freeness	Cannot prove to a third party how you voted	Anti-coercion: the system is designed so that a voter cannot produce convincing proof of their vote, even voluntarily

Receipt-freeness deserves elaboration because it is the property that prevents workplace coercion and vote-buying. In most voting systems, if an employer tells all employees "show me how you voted," employees can comply. The INDIS electoral design makes it cryptographically impossible for a voter to prove to a third party how they voted, even if they want to — because the verification receipt proves that a vote was cast, not which vote.

12.2 In-Person and Remote Voting

Both modalities are supported:

In-person voting (polling station):
- Citizen presents INDIS app at polling station
- App generates voter eligibility proof; polling station terminal verifies
- Citizen receives ballot (paper or electronic depending on station setup)
- Nullifier published on-chain to prevent double-voting
- Nullifier is not linkable to the citizen's identity

Remote voting (digital):
- Citizen casts ballot through app
- App generates encrypted ballot + voter eligibility ZK proof
- Ballot submitted; nullifier published
- Citizen receives cryptographic receipt to verify later inclusion

12.3 International Observer Access

A dedicated observer API allows accredited international observers to:

- Download all nullifiers (anonymized proof-of-vote records) for independent counting
- Run the public ZK verification algorithm against the electoral proof
- Query aggregate participation statistics with differential privacy
- Access the full audit log of system operations during the election window

Observer access is read-only and contains no citizen data. The same result that the Electoral Commission announces can be independently verified by any international observer with access to the public verification tools.

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13. Transitional Justice Module

13.1 Anonymous Testimony System

The anonymous testimony system is a specialized application of INDIS's ZK-proof capability to one of the most sensitive contexts of the transition: enabling witnesses and victims to contribute to the Truth Commission process without personal risk.

Design requirements that shaped the system:

- Witnesses must be verifiably Iranian (to prevent planted testimony from foreign intelligence operations)
- Witnesses must not be identifiable — even to the Commission, even under compulsion
- The same witness must be able to link multiple submissions without being identified
- If a witness chooses to identify themselves to a sealed court record, the system must facilitate this under strict controlled conditions

How it works:

A witness opens the testimony submission interface. They are not prompted for their name. They are prompted to generate a ZK citizenship proof — which proves they are an enrolled Iranian citizen, without revealing which one. They submit their testimony with this proof attached.

The Commission receives testimony + proof of citizenship. It cannot discover the identity of the witness. This is not a policy: if subpoenaed by any court in any jurisdiction, the system administrators cannot provide the witness's identity, because the system does not store it.

The witness receives a 256-bit random receipt token, stored locally on their device. Future submissions that include this token can be linked to the original by the Commission (same person, additional testimony) without identification.

13.2 Conditional Amnesty Workflow

The amnesty workflow has the opposite privacy model from testimony: applicants must be fully identified. This is required for victim notification and for the accountability component of the amnesty process.

The workflow:

1. Applicant enrolls in INDIS (Standard or Enhanced credential required — Social Attestation not sufficient for amnesty applications)
2. Applicant submits amnesty application with full identity disclosure, secured under additional encryption layer
3. Victim notification: victims associated with the case (identified through cross-referencing) receive notification that an amnesty application has been filed, without premature disclosure of applicant identity
4. Multi-party review committee receives application, with mandatory recusal checking against any committee member who has a conflict of interest
5. Decision communicated to applicant and, following defined process, to victims

Application data is stored under judicial multi-party escrow — no single person or organization can access it without the cooperation of all escrow holders.

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14. Government Portal — Ministry Operations

The Government Portal is the administrative interface through which ministry officials interact with INDIS. It is a secure web application, authenticated exclusively through X.509 certificates — no password-only access is permitted at any privilege level.

14.1 Dashboard and Monitoring

The portal provides a real-time dashboard with seven key operational metrics:

- Enrollment progress by province and population segment
- Credential issuance volume and type breakdown
- Pending review queue (social attestation cases; flagged enrollments)
- Revocation activity
- Verification volume by credential type
- System health indicators
- Audit event stream (sanitized; no PII)

14.2 Enrollment Review

Ministry officers review flagged enrollment cases — social attestation requests, document authenticity flags, biometric anomalies. The review interface shows:

- Enrollment data (documents, biometric confirmation status, attestor credentials)
- Fraud risk score (AI-generated)
- Recommended action with reasoning
- Comparable historical cases for reference

Review decisions (approve, reject, request additional documentation, escalate) are logged with the officer's credential and timestamp. Multi-level approval is required for operations affecting more than 1,000 records simultaneously.

14.3 Credential Issuance

Ministry operators can initiate credential issuance for specific populations — for example, the Electoral Commission issuing Voter Eligibility credentials before a referendum. The issuance workflow includes:

- Selection criteria for the target population
- Preview of affected citizens (count only; no PII without supervisor authorization)
- Multi-party approval for bulk issuance
- Scheduled issuance with rollback capability within a defined window

14.4 Bulk Operations and Audit Trail

Every action taken in the Government Portal generates an immutable audit entry. The audit log is:

- Hash-chained (each entry cryptographically links to the previous)
- Append-only at the database level (no deletion or modification possible)
- Queryable by time range, officer credential, action type, and affected population
- Retained for a minimum of 10 years
- Accessible (in sanitized form) to the Parliamentary Oversight Committee

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15. Offline and Low-Connectivity Access

15.1 The Rural Inclusion Imperative

Approximately 20–25 million Iranians live in areas with limited or no reliable internet connectivity. Their inclusion in the identity system is not optional — it is a political and humanitarian requirement. INDIS addresses this across three layers:

Layer 1 — USSD/SMS (feature phones, any connectivity level)

A national USSD short code provides basic credential status checks for citizens who do not own smartphones:

- `*ID#` — check enrollment status and basic credential validity
- `*ID*VOTE#` — check voter eligibility for current election cycle
- `*ID*PENSION#` — check pension payment eligibility
- SMS OTP — two-factor authentication for low-capability devices

USSD flows are available in Persian, Kurdish, Azerbaijani, and Arabic. No citizen data is retained after session end — sessions are stateless by design.

Layer 2 — Mobile enrollment units (full enrollment, satellite-connected)

Mobile enrollment units are self-contained vehicles carrying biometric hardware, satellite connectivity, a generator, and two trained agents. They can conduct full enrollment — including biometric capture, document processing, and credential issuance — in areas with no ground infrastructure.

Layer 3 — Offline-capable mobile app (ZK proof generation without network)

The INDIS mobile app generates ZK proofs entirely on-device, without network connectivity. A citizen in a remote area can prove their voter eligibility, age, or credential validity to an offline verifier terminal — with the verifier using a cached revocation list (valid for up to 72 hours).

This covers the scenario of a polling station in a village with no connectivity on election day: the station terminal operates on its 72-hour cached revocation list, citizens generate proofs on their phones, and verification proceeds completely offline.

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16. Physical Identity Card

The physical identity card is designed for citizens who do not own smartphones and for use cases (border crossing, medical emergencies) where a physical credential is preferable or required.

Technical specifications:

- Standard: ICAO 9303 Machine Readable Travel Document
- Contact chip: ISO 7816
- NFC interface: ISO 14443 (contactless)
- Chip stores: DID document, public key, validity status
- Chip does NOT store: biometric templates, national ID in plaintext, address
- Visual display: Persian name (Nastaliq script), masked national ID, photograph, validity date, QR code

- Offline verification: any card-reader terminal with cached NIA public key can verify the card without network access

First issuance: Free for all citizens. This is a political commitment — identity must not be a service for those who can pay.

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17. Pension and Subsidy Integrity

The pension integrity use case is one of the most immediately financially impactful applications of INDIS. The mechanism is straightforward:

1. All pension-age citizens are enrolled in INDIS through a targeted enrollment campaign (mobile units to rural areas; fixed centers for urban)
2. Pension payment processing is linked to credential validity — payment requires a valid, non-expired Pension credential
3. Pension credentials expire monthly and can only be renewed through biometric re-verification (in-person at center or mobile unit, or via the app for smartphone users)
4. A deceased citizen's credential cannot be renewed — no one else shares their biometric profile
5. Ghost beneficiaries who cannot produce biometric verification have their payments suspended after a defined grace period, with an appeal process for legitimate edge cases (e.g., hospitalized citizens)

The financial impact: based on comparable programs in other countries (Nigeria's GIFMIS, Ghana's pension audit, Kenya's Huduma Namba), estimated recovery of 5–8% of the pension budget within 90 days of full enrollment, translating to several hundred billion rials per year.

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18. Privacy Control Center

The Privacy Control Center is a prominently accessible feature of the citizen app — not buried in settings menus. Its placement reflects a political commitment: citizens are in control of their own identity.

What citizens can see:

- Complete history of every verification request received, with verifier category, date, and credential type
- List of every organization authorized to request verification from them
- Consent records for every credential sharing event
- Data minimization settings by verifier category

What citizens can do:

- Set per-verifier rules: Always Share / Always Ask / Never Share
- Receive real-time notifications for verification attempts
- Decline any verification request in real time
- Request a complete data export (cryptographically signed, delivered within 72 hours)
- Submit a data correction request (tracked, with defined resolution timeline)
- Escalate unresolved complaints to the independent Identity Ombudsman

What citizens cannot prevent:

- Level 4 emergency override (requires senior multi-party authorization; generates automatic Parliamentary Committee alert; logged permanently)
- Judicial disclosure under a valid judicial order (requires judicial multi-party escrow process)

The Privacy Control Center is the citizen-facing expression of the system's fundamental commitment: the government is the credential issuer, not the data owner. Citizens own their identity.

PART IV: HOW THE SYSTEM WORKS

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19. System Architecture

19.1 High-Level Architecture

INDIS is organized in four layers:

```
┌──────────────────────────────────────────────────────────────────────┐
│                        CITIZEN LAYER                                 │
│   Android App  │  iOS App  │  HarmonyOS App  │  PWA  │  USSD/SMS   │
│   Kiosk Terminal  │  Verifier Terminal  │  Physical Card            │
└──────────────────────────────┬──────────────────────────────────────┘
                               │ HTTPS / mTLS
┌──────────────────────────────┴──────────────────────────────────────┐
│                        API GATEWAY  (:8080)                          │
│  JWT Auth  │  mTLS  │  WAF  │  Rate Limiting  │  Circuit Breaker    │
└──────────────────────────────┬──────────────────────────────────────┘
                               │ gRPC (internal)
┌──────────────────────────────┴──────────────────────────────────────┐
│                     CORE SERVICES LAYER (Go)                         │
│                                                                       │
│  identity  :9100   credential  :9102   enrollment  :9103             │
│  biometric :9104   audit       :9105   notification :9106            │
│  electoral :9107   justice     :9108   verifier    :9110             │
│  govportal :8200   ussd        :8300   card        :8400             │
│  gateway   :8080                                                      │
│                                                                       │
│  zkproof (Rust) :8088    ─── Groth16 / STARK / Bulletproofs          │
│  ai (Python) :8000       ─── Biometric dedup / fraud detection       │
└──────────────────────────────┬──────────────────────────────────────┘
                               │
┌──────────────────────────────┴──────────────────────────────────────┐
│                         DATA LAYER                                   │
│  PostgreSQL 16 (primary)   │  Redis 7 (session / revocation cache)  │
│  Kafka (event streaming)   │  Hyperledger Fabric (DID/anchor/audit) │
│  HashiCorp Vault + HSM (key management — FIPS 140-2 Level 3)         │
└──────────────────────────────────────────────────────────────────────┘
```

19.2 Design Principles Embedded in the Architecture

Separation of identity from data. The identity service knows your DID. The credential service issues your credentials. The biometric service holds your template. No single service holds a complete picture of who you are. Cross-service correlation requires multi-party authorization and leaves an audit trail.

Layered communication security. All internal service-to-service communication uses mutual TLS (mTLS) — both sides authenticate each other with certificates. A compromised external connection cannot impersonate a core service. A compromised core service cannot impersonate another.

Everything is auditable. OpenTelemetry distributed tracing is active across all 15 services. Every request that enters the system is traceable from the API gateway through every service it touches. Prometheus metrics provide real-time operational visibility. All of this runs on sovereign infrastructure — no telemetry is sent to foreign services.

Defense in depth. Security is not a perimeter — it is layers. The WAF blocks known attack patterns at the gateway. Rate limiting prevents brute-force attacks. JWT claims are validated at the service layer independently of the gateway. Sensitive operations require multi-party authorization even for administrators.

20. Zero-Knowledge Proof Engine

The ZK proof engine is a purpose-built Rust service that implements four ZK proof systems, each selected for a specific use case.

20.1 Why Four Proof Systems?

No single ZK proof system is optimal for all use cases. The choice of proof system involves tradeoffs between proof size, verification time, generation time, trusted setup requirements, and post-quantum security.

Proof System	Used For	Key Advantage	Tradeoff
Groth16 (arkworks)	Standard credential verification	Smallest proofs; fastest verification; fast phone generation	Requires trusted setup ceremony
Winterfell STARK	Electoral / referendum verification	Post-quantum secure; no trusted setup; publicly verifiable	Larger proof size; slower generation
PLONK	Batch credential operations	Universal trusted setup; good for complex statements	Medium proof size; medium speed
Bulletproofs	Anonymous testimony; range proofs	No trusted setup; compact range proofs	Slower verification for complex proofs

20.2 Proof Generation Architecture

ZK proofs in INDIS are generated on two surfaces:

On-device (mobile app and Citizen PWA): For real-time verification at service points, the ZK circuit and proving key are pre-loaded on the device. Proof generation happens locally, in under 5 seconds on a 2022 mid-range Android phone. No server involved — no network required.

Server-side (ZK proof service, :8088): For electoral ballots and other operations where the proof statement is complex, or where the device does not meet minimum performance requirements, proof generation is delegated to the server-side proof engine. Server-side generation uses hardware acceleration and completes in under 1 second.

20.3 ZK Circuits

The circuits — the mathematical programs that define what statements can be proven — are written in Circom 2.0 and cover:

- `age_proof.circom` — proves age ≥ threshold without revealing exact age

- `citizenship_proof.circom` — proves Iranian citizenship without revealing identifier
- `voter_eligibility.circom` — proves citizenship + age ≥ 18 + not in exclusion list, atomically
- `credential_validity.circom` — proves credential issued by authorized issuer + not revoked + not expired
- `electoral_proof.cairo` — Cairo circuit for STARK-based electoral proof generation

All circuits are open-source and will be formally verified (mathematical proof that the circuit is sound and complete) before production deployment. The formal verification process uses Ecne (Columbia University) or Picus (Princeton/NYU), and the verification reports are published publicly.

20.4 The Trusted Setup Ceremony

Groth16 and PLONK require a one-time trusted setup ceremony — a multi-party computation in which multiple independent parties each contribute a random secret, and the final proving key is computed from all contributions combined. The security guarantee is: if even one participant destroys their secret after contributing, the resulting key is secure and no one can forge proofs.

For INDIS, the trusted setup ceremony will:
- Involve a minimum of 20 participants from at least 5 countries
- Include diaspora representatives as well as domestic participants
- Be conducted in public, with all participants and their contributions publicly logged
- Be independently verified by international cryptographers
- Be recorded and published in full

---

21. Cryptographic Infrastructure

21.1 Algorithm Selection

Context	Algorithm	Standard	Rationale
Digital signatures (current)	Ed25519	RFC 8032	Fast; compact; widely audited
Digital signatures (long-term credentials)	CRYSTALS-Dilithium3	NIST FIPS 204	Post-quantum; long-term security
Key agreement	ECDH (P-256)	NIST SP 800-56A	Widely supported; well-audited
Symmetric encryption	AES-256-GCM	NIST SP 800-38D	Gold standard for data at rest

Context	Algorithm	Standard	Rationale
Hash functions	SHA-256, SHA-3-256	NIST FIPS 180-4	Blockchain anchoring; deduplication
Transport security	TLS 1.3	RFC 8446	Current standard; forward secrecy

21.2 Key Management and HSM

All cryptographic keys used by the government are managed through HashiCorp Vault with a FIPS 140-2 Level 3 HSM backend. Key management principles:

- **Government issuer keys never leave the HSM.** Signing operations are performed inside the HSM; the private key is never exported to application memory.
- **Citizen private keys never leave the device.** There is no concept of "key escrow" for citizen keys in the system design.
- **Key rotation** is automated and audited. Every rotation generates an audit event.
- **Key ceremonies** for root keys require multi-party authorization — no single NIA administrator can perform a key ceremony alone.

21.3 Post-Quantum Readiness

The `pqc-migrate` tool is included with the system. When quantum computing reaches the threshold that threatens Ed25519 (estimated 10–15 years away under current projections), the system can:

1. Run `pqc-migrate` to re-sign all existing credentials with Dilithium3 in batches
2. Enable `--pqc-mode` on the credential service to issue new credentials with Dilithium3 signatures
3. Complete migration without downtime or re-enrollment

Long-term credentials (those valid for more than 2 years) are issued with Dilithium3 from launch. Short-term credentials use Ed25519 until the migration is triggered.

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22. Blockchain Anchoring Layer

22.1 What the Blockchain Is and Is Not

The blockchain is not a database. It is a tamper-evident ledger of hashes and status flags. The personal information that makes up an Iranian citizen's identity is never written to the blockchain.

The blockchain stores:

- DID Documents (public keys and service endpoints — no personal data)
- Credential hash anchors (a fingerprint of the credential, not its contents)
- Revocation status flags (boolean: revoked or not)
- Anonymized verification event records
- Electoral nullifiers (proof-of-vote without voter identity)

The blockchain does not store:

- Names, addresses, national ID numbers
- Biometric data
- Credential contents
- Any personally identifiable information

This separation is enforced at the chaincode (smart contract) level. The chaincode actively rejects any transaction that contains data matching personal information patterns. A malicious administrator cannot write personal data to the blockchain — the code prevents it.

22.2 Hyperledger Fabric Configuration

The system is implemented on Hyperledger Fabric — a permissioned blockchain platform that allows the Iranian government to operate a sovereign ledger with controlled membership and no dependency on any foreign validator.

```
Network: 21+ peer nodes across 4 organizations
  Organization 1: NIA (6 peers — Tehran, Isfahan, Mashhad, Tabriz, Ahvaz, Shiraz)
  Organization 2: Ministry of Interior (5 peers)
  Organization 3: Ministry of Health (5 peers)
  Organization 4: Ministry of Finance (5 peers)

Orderers: 4 (Raft consensus) — Tehran, Isfahan, Mashhad, Tabriz

Channels:
  did-registry-channel    — DID documents (all organizations)
  credential-anchor-channel — Hash anchors + revocation (all organizations)
  audit-log-channel       — Verification events (all organizations)
  electoral-channel       — Electoral nullifiers (NIA + Electoral Commission)

Endorsement Policy:
  Credential operations: 3-of-5 NIA peers
  DID registration: 2-of-3 NIA peers
  Revocation: NIA + issuing organization (2-of-2)
```

22.3 Blockchain Abstraction

The application layer communicates with the blockchain only through a well-defined abstraction interface. This means:

- If Hyperledger Fabric needs to be replaced (for performance, governance, or sovereignty reasons), the replacement requires only a new adapter implementation — no changes to the 15 core services
- Different blockchain implementations can be tested in parallel environments
- The system can operate with a `MockAdapter` (for development and testing) or a `FabricAdapter` (for production) with no code changes in core services

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23. Biometric Deduplication Engine

23.1 Architecture

The biometric engine is a Python service (FastAPI, :8000) that bridges the enrollment pipeline and the biometric database. It provides:

- **Face recognition:** CNN-based embedding (FaceNet/ArcFace architecture) producing a 512-dimensional face vector. Similarity scored by cosine distance.
- **Fingerprint matching:** NIST BOZORTH3-compatible minutiae extraction and matching. 1:N matching against enrolled database.
- **Multi-modal fusion:** Combined similarity score from face + fingerprint, weighted by capture quality. Flags for human review when confidence is below threshold.
- **Liveness detection:** ISO/IEC 30107-3 compliant presentation attack detection. Prevents use of photographs, videos, or masks to defeat the biometric check.

23.2 Privacy Guarantees

The biometric engine is the most privacy-sensitive component of the system. Its privacy properties:

- Biometric templates are never stored in plaintext — AES-256-GCM encryption, HSM-managed keys
- One-way transformation applied before storage: original biometric cannot be reconstructed from stored template
- Biometric data is physically isolated on a separate database server with no external network access; accessible only from the biometric service
- Biometric data is **never shared** with any foreign government, organization, or private company — this is a hard architectural constraint, not a policy

- Alternative biometric pathways for persons with disabilities (occupational fingerprint wear, visual impairment)

23.3 Deduplication at Scale

For a population of 88 million, comparing a new enrollment against every existing record is computationally intensive. The system uses:

- Locality-sensitive hashing (LSH) to partition the comparison space, reducing from 88M comparisons to ~10,000 per enrollment
- GPU-accelerated batch processing for high-enrollment-rate periods (up to 500,000 per day)
- Confidence-stratified output: definite match (auto-flag), probable match (human review), clear (proceed)

---

24. Data Architecture and Privacy Guarantees

24.1 What Data Lives Where

Data Type	Storage Location	Access Control	Retention
DID Documents + public keys	Hyperledger Fabric (blockchain)	Public read; NIA-authorized write	Permanent
Credential hash anchors	Hyperledger Fabric (blockchain)	Public read; NIA-authorized write	Permanent
Citizen biometric templates	Encrypted biometric DB (air-gapped)	Biometric service only; no external access	25 years post-death
Citizen credentials	Citizen's device (encrypted wallet)	Citizen's private key only	Citizen-controlled
Enrollment record (documents, photo)	PostgreSQL (encrypted, HSM keys)	NIA — specific access by role	25 years
Audit log	PostgreSQL (append-only) + Fabric	Append-only; read by authorized auditors	10 years minimum
Session data	Redis (TTL-based)	Gateway service	Session duration only
Anonymous testimony	Encrypted PostgreSQL	Judicial escrow; Commission (anonymized)	50 years

24.2 Cross-Verifier Correlation Prevention

One of the most important privacy guarantees of INDIS is that verifiers cannot build behavioral profiles of citizens by comparing notes. This is a known risk in digital identity systems: if a citizen uses the same identifier at a pharmacy, a bank, and a government office, those three organizations can correlate their data to build a profile.

INDIS prevents this architecturally:

- Each ZK proof includes a verifier-specific nonce — the same credential generates a different-looking proof for each verifier
- The citizen's DID is not revealed to Level 1 verifiers — they receive only the proof, not the identifier
- The blockchain records anonymized verification events without verifier-linkable identifiers

A verifier cannot determine whether the person who used their service yesterday also used a competing service. This is not a policy claim — it is a cryptographic property.

24.3 Differential Privacy for Statistics

Aggregate statistics (how many people in Isfahan are enrolled; what percentage of the 31–50 age bracket has Voter Eligibility credentials) are computed using differential privacy techniques, with the privacy parameter (ε) published publicly. This means:

- No individual's data can be reverse-engineered from published statistics
- The government can publish accurate population-level data for policy purposes
- Individual privacy is preserved mathematically

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25. Government Observability, Data Sharing, and Authorized Access

A common and legitimate concern raised by policy makers and ministry officials is whether a privacy-preserving identity system can still support the operational needs of government. The answer is yes — but it requires understanding precisely what the ZK privacy layer protects, what it does not protect, and the separate, well-defined channels through which the government accesses the data it genuinely needs.

The ZK architecture does not make INDIS opaque to government. It makes INDIS opaque to the wrong parties at the wrong moments. The distinction is critical.

25.1 What ZK Proofs Actually Protect — and What They Do Not

The zero-knowledge proof layer operates exclusively at the **citizen-to-verifier boundary**. It governs what a pharmacy, a bank, a border checkpoint, or an employer learns when a citizen presents a credential. In that interaction, the verifier receives a boolean result and nothing else.

The ZK layer has no bearing on:

- What the government records at enrollment
- What the government can query in its own authorized databases
- What audit information is generated by system operations
- What data ministries can access within their authorized scope
- What investigators can obtain under judicial authorization
- What the Parliamentary Oversight Committee can review
- What aggregate statistics the government can compute and publish

The following table clarifies the boundary precisely:

Interaction	What ZK Protects	What Remains Visible
Citizen proves age at pharmacy	Verifier sees only boolean	NIA: anonymized verification event; citizen: entry in privacy log
Citizen votes in referendum	Polling station sees only eligibility	Electoral Commission: total participation count, nullifier set
Ministry of Finance checks pension eligibility	Not applicable — Finance is the credential issuer	Finance: full pension record for that citizen, within authorized scope
Border guard verifies identity at Level 2	Identity attributes not transmitted unless citizen consents	Border Authority: credential category + validity date; NIA audit log records the event
Judicial investigation under court order	ZK layer bypassed under Level 3 disclosure	Court: full verified identity; audit committee notified; event logged permanently
System administrator queries enrollment database	No ZK involved — administrative access	Full audit trail generated; multi-party authorization required; Parliamentary Committee alerted

25.2 What the Government Always Sees

The following data is always available to the government within defined access controls, regardless of the ZK layer:

Enrollment records — The NIA holds complete enrollment data for every citizen: documents submitted, biometric confirmation status, enrollment pathway, enrollment date, and the processing agent. This is the authoritative civil record. Ministry of Interior can query it within their authorized scope.

Credential issuance log — Every credential issued — to whom (by DID), by which authority, when, and under what parameters — is logged immutably. The issuing ministry always has full visibility into what it issued.

Aggregate operational statistics — Real-time dashboards show total enrollment by province, demographic bracket, credential type, and period. These statistics use differential privacy to protect individuals while giving government the population-level data it needs for policy and planning.

Audit event stream — Every system operation — credential issuance, revocation, role assignment, bulk operation, administrative access, Level 3 or Level 4 override — generates an immutable, timestamped, cryptographically signed audit entry. This stream is available to authorized auditors and the Parliamentary Oversight Committee.

Revocation and fraud signals — The biometric deduplication engine flags suspicious enrollment patterns. The system generates anomaly alerts for unusual agent behavior, duplicate biometric attempts, and velocity anomalies (e.g., unusually high social attestation volumes from a single region or agent). These feed directly into the Ministry of Interior's security dashboard.

25.3 Tiered Ministry Data Rights

Not all government access is equal, and the system enforces this structurally. Each ministry is issued a certificate that defines exactly what it can query. This is not a policy setting that an administrator can override — it is enforced at the service layer, independently of the Government Portal.

Ministry / Authority	Authorized to Query	Not Authorized to Query
Ministry of Interior	Enrollment records; citizenship; residency; voter rolls	Health records; pension amounts; security clearance basis
Ministry of Finance	Pension credential status; tax identity linkage	Health history; voting behavior; security clearance
Ministry of Health	Health insurance enrollment; pharmacy dispensing eligibility	Pension amounts; voting status; security clearance
Electoral Commission	Voter eligibility; nullifier set; participation counts	Health; pension; employment records
Ministry of Justice	Full identity under judicial order (Level 3, logged)	Records outside the specific tribunal's authorization
NISS / Military Command	Security clearance status; enrollment confirmation	Pension; health; voting behavior
NIA System Administrators	System configuration; key ceremonies; audit log	Individual citizen records without multi-party authorization

Access outside a ministry's authorized scope generates an automatic access-denied event and alerts the Parliamentary Oversight Committee.

25.4 Authorized Data Sharing Between Ministries

Inter-ministry data sharing is a legitimate and frequent operational need. INDIS supports this through **credential-mediated sharing**: instead of raw database queries between ministries, each ministry queries whether a citizen holds a specific credential issued by another. The querying ministry learns only what the credential asserts — not the underlying record.

For cases where the credential alone is insufficient — judicial proceedings, cross-ministry fraud investigations, national security matters — a formal data sharing request is initiated:

1. The requesting authority submits a request specifying: requester credential, subject DID, data requested, legal basis, and approving official's credential
2. NIA's data governance function reviews the request with defined response timelines
3. If approved, the specific data is released under an audited, time-limited access grant
4. The citizen is notified of the access event (deferred in active security investigations pending judicial authorization)
5. The full record — request, approval, access — is retained in the immutable audit log

25.5 Judicial and Investigation Access

INDIS is not a barrier to lawful investigation. It is a framework that makes investigation access **accountable rather than invisible**.

Level 3 — Full Identity Disclosure (judicial authorization)

1. Court order submitted to NIA with the issuing judge's cryptographic signature
2. NIA data governance reviews for validity (jurisdiction, scope, expiry)
3. Full identity data released to the requesting court or investigator
4. Subject citizen notified (unless the order specifically prohibits notification, in which case a sealed notification is held)
5. Parliamentary Oversight Committee receives an anonymized count in its monthly report
6. Event logged permanently in the immutable audit trail

Level 4 — Emergency Override (senior multi-party authorization)

1. Two senior NIA officers with separate HSM-backed certificates must authorize simultaneously
2. Override logged immediately and automatically in the audit trail
3. Real-time alert sent to the Parliamentary Oversight Committee
4. Override expires after a defined window (maximum 48 hours); continuation requires judicial authorization

5. Post-event review is mandatory; the oversight committee may demand a formal explanation

This means emergency access is fast and operationally viable — but it is visible, bounded, and reviewed. It cannot be used silently or repeatedly without accountability.

25.6 The Audit Trail as a Government Asset

The immutable audit log is not only a privacy protection for citizens — it is one of the most valuable operational assets the government holds.

Accountability for officials: Every action by every ministry official — enrollment approval, credential issuance, data access, role assignment — is recorded with the official's credential, timestamp, and the resource accessed. When questions arise about administrative decisions, the audit trail is the authoritative record.

Fraud detection: Patterns of unusual access, bulk operations outside normal parameters, repeated overrides by the same official, or access to records with no operational justification are all detectable from the audit stream.

Parliamentary oversight and public trust: The government can publish aggregate audit statistics — Level 3 disclosures per month, Level 4 overrides per quarter, credential issuances by type — as a demonstration of accountable governance. In a transition context, this is a political asset.

Legal evidence: In disputes about administrative decisions, the audit log provides an authoritative, tamper-evident record of what happened, when, and who authorized it.

International credibility: International partners — particularly for AML/KYC compliance, electoral observation, and development finance — require demonstrated accountability. A cryptographically sound operational record provides it.

25.7 Aggregate Data and Policy Analytics

The government needs population-level data for policy planning: enrollment progress by province, pension-age verification rates, regional distribution of Social Attestation credentials (a proxy for historical exclusion requiring targeted remediation).

INDIS provides this through a dedicated analytics layer:

Differential privacy: Statistical queries are answered with calibrated noise such that no individual record can be reverse-engineered from the published statistic. The privacy parameter (epsilon) is published alongside each report.

Real-time dashboards: The Government Portal shows enrollment progress, verification volumes, and credential issuance rates — at the population level, with individual citizen data only accessible under authorized workflows.

Policy reporting: Monthly population reports by province, age bracket, credential type, and enrollment pathway — shareable with international partners and publishable as accountability documents.

25.8 Summary: Supported and Prevented

Government Need	Supported?	Mechanism
Know who is enrolled; verify enrollment records	Yes	Enrollment database; authorized ministry access
Issue and manage credentials	Yes	Credential service; government portal
Verify a citizen's identity for service delivery	Yes, all four levels	Level 1 ZK boolean; Level 2 NFC; Level 3 full identity; Level 4 emergency override
Access citizen data for a ministry's own functions	Yes, within scope	Role-based access; credential-mediated inter-ministry queries
Conduct a lawful investigation	Yes, with authorization	Level 3 judicial disclosure; permanent audit trail
Execute an emergency override	Yes, with accountability	Level 4 multi-party authorization; automatic oversight alert
Track aggregate enrollment and service usage	Yes	Differential-privacy statistics; policy dashboards
Audit pension beneficiary liveness	Yes	Pension credential requires biometric re-verification to renew
Detect enrollment or credential fraud	Yes	Biometric deduplication; anomaly detection; audit log analysis
Provide evidence in administrative or judicial proceedings	Yes	Immutable audit trail; certified credential records
Track which shops or services a citizen uses across verifiers	No	ZK proofs reveal nothing to verifiers; cross-verifier correlation is architecturally impossible
Build a behavioral profile without judicial authorization	No	No behavioral data is generated; the system produces boolean results only at verifier boundaries
Access another ministry's records without authorization	No	Service-layer access controls; unauthorized attempts generate automatic oversight alert

The final three rows are the only things the ZK architecture prevents. Everything above them is fully supported, documented, and operationally straightforward. The purpose of the privacy architecture is not to impede government — it is to ensure that when government accesses citizen data, it does so through a defined, accountable, and auditable process rather than through invisible administrative convenience.

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26. Security Architecture and Threat Model

26.1 Threat Landscape

INDIS is designed assuming an active, sophisticated adversary. The threat model includes:

Threat Actor	Capability	Specific Attack Scenarios
Regime remnants	Insider access; physical presence; corrupt agents	Fraudulent identity creation for persons avoiding accountability; credential forging; enrollment agent bribery
Foreign intelligence agencies	Nation-state cyberattack capability; supply chain attacks	Electoral infrastructure compromise; biometric database exfiltration; proof system cryptanalysis
Organized fraud rings	Coordinated enrollment fraud; ghost beneficiary maintenance	Multiple-identity enrollment; biometric spoofing; pension fraud at scale
Future authoritarian government	Full institutional access; legal compulsion	Attempting to use INDIS for surveillance; forcing credential-linked tracking
Insider threats	Privileged access; legitimate credentials	Data exfiltration; unauthorized credential issuance; audit log manipulation

25.2 Mitigations by Threat

Against regime remnants:

- All agent credentials are cryptographically bound to the agent's own biometric — cannot be transferred
- Social attestation requires three enrolled attestors, each with an immutable cryptographic signature — forging is detectable
- Anomaly detection flags unusual enrollment patterns (e.g., a single agent enrolling 50 social attestation cases in one day)
- Supervisory review required for all social attestation decisions

Against foreign intelligence:

- All infrastructure is sovereign — no foreign cloud services, no foreign software with privileged access
- Air-gapped biometric database — not reachable from the internet
- ZK proof systems are open-source and formally verified — no backdoors possible in the cryptographic layer
- Regular red team exercises with international participation (scenarios include nation-state)

Against future authoritarian governments:

- The ZK architecture makes mass surveillance technically impossible at the protocol level — there is nothing to surveil
- Citizen private keys cannot be compelled — the government never had them
- Parliamentary Oversight Committee has automatic alert on any Level 4 override
- The code is open-source — any attempt to introduce surveillance capability would be publicly visible
- Multi-party key management means no single individual or entity can access root keys alone

Against insider threats:

- Role-based access control at system level — not overridable by individual administrators
- All privileged operations require multi-party authorization
- Comprehensive audit log that cannot be modified or deleted
- Separation of duties enforced in the architecture — the key custodian is not the same as the enrollment approver

25.3 Security Testing Requirements

Before any production deployment:

- Minimum 2 independent security audits by internationally recognized firms
- Formal verification of all ZK circuits (Ecne or Picus)
- Penetration testing of all user-facing surfaces
- Red team exercise simulating regime-remnant insider attack
- Red team exercise simulating nation-state cyberattack on electoral infrastructure
- Public bug bounty program established and running for minimum 30 days before Phase 2 (electoral) launch
- Load testing at 5× peak for electoral module; results published publicly

26. Technology Stack

26.1 Core Technologies

Layer	Technology	Rationale
Backend services	Go 1.22	Performance-critical concurrent services; single binary deployment; growing Iranian expertise
ZK proof engine	Rust 1.75	Memory safety eliminates entire classes of cryptographic vulnerabilities; best-in-class ZK library ecosystem
AI/biometric service	Python 3.11 + PyTorch	State-of-the-art biometric ML ecosystem; ONNX interoperability for on-device inference
Mobile — Android	Kotlin + Jetpack Compose	Primary Iranian mobile platform (~65% market share)
Mobile — iOS	Swift + SwiftUI	Secondary platform; high diaspora usage
Mobile — HarmonyOS	ArkTS + ArkUI	Required for Huawei devices common in Iran under sanctions
Web clients	React 18 + TypeScript + Vite	RTL-capable; PWA support; large developer community
Internal APIs	gRPC	Type-safe; efficient binary protocol; strong service contract enforcement
External APIs	REST + OpenAPI 3.0	Universal compatibility; machine-readable spec for client codegen
Government portal	GraphQL (Apollo)	Flexible queries for complex ministry reporting needs
Primary database	PostgreSQL 16	Battle-tested; full-featured; open-source; excellent encryption support
Session/cache	Redis 7	Fast revocation cache; session management
Event streaming	Apache Kafka	Reliable async event chain (enrollment → credential → audit → notification)
Key management	HashiCorp Vault + FIPS 140-2 L3 HSM	Industry standard; HSM integration; auditable key operations
Blockchain	Hyperledger Fabric	

Layer	Technology	Rationale
		Permissioned; sovereign; Go chaincode; no external validator dependency
Infrastructure	Kubernetes + Helm + Terraform	Reproducible infrastructure; GitOps-compatible
Observability	Prometheus + Grafana + OpenTelemetry	Full-stack metrics + tracing; no foreign telemetry
CI/CD	Self-hosted GitLab	No foreign service dependency; full sovereign control of pipeline
ZK circuits	Circom 2.0 + arkworks (Rust) + Winterfell (Rust)	Mature ecosystem; formally verifiable; open-source
Typography	Vazirmatn	Open-source; complete Persian Unicode; optimized for mobile; Iranian-community maintained

26.2 What Is Explicitly Excluded

Category	Excluded	Reason
Cloud platforms	AWS, Azure, Google Cloud	Foreign government subpoena risk; sovereignty violation
Proprietary cryptography	Any closed-source crypto library	Cannot be audited; backdoor risk cannot be assessed
Public blockchains	Ethereum, Solana, Bitcoin	No sovereignty; foreign validator control; cost unpredictability
Foreign biometric SDKs	Any SDK without full source audit rights	Cannot verify absence of data exfiltration or backdoors
Third-party analytics	Google Analytics, Mixpanel, etc.	Behavioral data would leave sovereign control

27. Performance and Availability Requirements

27.1 Response Time Targets

Operation	Target	Maximum	Measurement Context
ZK proof generation (Groth16, on-device)	2 seconds	5 seconds	2022 mid-range Android
ZK proof generation (STARK, electoral)	5 seconds	15 seconds	Server-side
Proof verification at terminal	200ms	500ms	Online; online revocation check
Proof verification at terminal	100ms	300ms	Offline; cached revocation list
Biometric deduplication	30 seconds	90 seconds	Full 88M population
Credential issuance	5 seconds	30 seconds	Standard pathway
App cold start to ready	2 seconds	5 seconds	—
Blockchain write finality	1 second	3 seconds	Hyperledger Fabric Raft
Revocation propagation	30 seconds	60 seconds	Online verifier nodes

27.2 Availability Requirements

Service	SLA	Max Downtime/ Year	Recovery Point	Recovery Time
Core identity verification	99.99%	53 minutes	15 minutes	1 hour
Enrollment services	99.9%	8.7 hours	1 hour	4 hours
Electoral services (active election window)	99.999%	5 minutes	5 minutes	30 minutes
Government portal	99.9%	8.7 hours	1 hour	4 hours
Non-critical reporting	99.5%	44 hours	4 hours	8 hours

27.3 Scale Targets by Phase

Phase	Enrolled Population	Peak Verification Rate	Enrollment Rate
Phase 1 (Month 3)	2 million	100,000 / hour	500,000 / day
Phase 2 (Month 4 — referendum)	10 million	2,000,000 / hour (election day)	100,000 / day
Phase 3 (Month 12)	50 million	500,000 / hour	200,000 / day
Phase 4 (Month 24)	88 million	1,000,000 / hour	50,000 / day (maintenance)

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28. Infrastructure and Operations

28.1 Deployment Architecture

INDIS runs on a Kubernetes cluster deployed on sovereign Iranian infrastructure — no cloud services, no data leaving Iranian jurisdiction. The deployment is fully described as code (Helm charts for all 15 services + infrastructure, Terraform for provisioning), which means:

- Reproducible deployments — every deployment is identical by construction
- GitOps workflow — infrastructure changes are reviewed, approved, and audited before application
- Disaster recovery — a complete new cluster can be provisioned from code in hours

28.2 Geographic Distribution

For resilience, the system is deployed across a minimum of three geographic data centers:

- **Primary:** Tehran — main operations, all services active
- **Secondary:** Isfahan — hot standby; all services running; can take over within minutes
- **Tertiary:** Mashhad or Tabriz — warm standby; can activate within 1 hour

28.3 Observability

Every service exposes:

- **Prometheus metrics** (request rates, error rates, latency percentiles, business-level counters)
- **OpenTelemetry traces** (distributed trace for every request across all services)
- **Structured logs** (sanitized — no PII in logs; correlated with traces)

A Grafana dashboard provides real-time operational visibility. Alert rules are configured for all SLA-impacting conditions. The entire observability stack runs on sovereign infrastructure.

PART V: GOVERNANCE AND DEPLOYMENT

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29. Deployment Roadmap — Phased by Transition Timeline

The deployment roadmap is structured around the Emergency Phase Booklet's requirements, with hard deadlines for the highest-priority political deliverables.

```
Day:    1    15   30   40   60   90        Month 4    Month 12   Month 24
        |    |    |    |    |    |              |          |          |
        ├────┤                                                        |
PHASE 0 |────| Foundation, governance, HSM deploy, staging audit      |
        |                                                             |
        ├──────────────────────┤                                      |
PHASE 1               ──────────| Priority personnel vetting         |
                                |  Day 40: Security clearance live   |
        |                                                             |
                 ├──────────────────────────┤                         |
PHASE 2          |        ────────────────────| Electoral + Justice    |
                 |  Day 60: Testimony live   |                        |
                 |  Month 4: Referendum ready|                        |
                 |                                                    |
                         ├──────────────────────────────────┤         |
PHASE 3                  |      ──────────────────────────────| National  |
                         |                                   | rollout   |
                                                             |          |
                                                  ├──────────────────────────┤
PHASE 4                                           |              Full coverage  |
```

29.1 Phase 0 — Foundation (Days 1–30)

Governance:

- Establish National Identity Authority (NIA) as governing body with defined mandate
- Publish Data Protection Impact Assessment for Phase 1
- Establish Parliamentary Oversight Committee with technical staff
- Initiate blockchain platform deployment (Hyperledger Fabric network provisioning)
- Appoint independent auditors for Phase 1 security review

Technical:

- Deploy production infrastructure across three geographic locations
- Establish HSM cluster and conduct root key ceremony (multi-party, international witnesses)
- Deploy all 15 core services in production configuration
- Deploy CI/CD pipeline (self-hosted GitLab)
- Complete penetration testing of Phase 1 scope
- Deploy monitoring and observability stack
- Publish public bug bounty program

Deliverables:

- Production environment passing security audit
- Governance framework operational and published
- Public documentation of system architecture (this document and accompanying technical specs)

29.2 Phase 1 — Priority Personnel Enrollment (Days 15–40)

Target: Military, security services, civil servants, healthcare workers — approximately 2 million people.

Hard Deadline: Day 40 — all priority military and security personnel enrolled and security clearance credentials issued (Emergency Phase Booklet — Military Chapter).

Milestones:

- Mobile apps (Android, iOS, HarmonyOS) and enrollment agent applications deployed
- 50+ fixed enrollment centers operational (provincial capitals and major cities)
- Bulk enrollment pipeline operational for institutional batch processing
- Initial credential types live: Citizenship, Age Range, Security Clearance
- Government portal operational for Ministry of Interior and Ministry of Defense
- USSD gateway live for basic status checks

Scale operations:

- Bulk enrollment can process 100,000 personnel per day at 50 centers
- 2 million enrollments achievable within 20 working days
- Security Clearance credential issuance is same-day after biometric capture

29.3 Phase 2 — Electoral Preparation and Justice Infrastructure (Days 30–Month 4)

Hard Deadline: Day 60 — Anonymous testimony system operational (Truth Commission, Emergency Phase Booklet).

Hard Deadline: Month 4 — Electoral module independently audited and referendum-ready.

Milestones — Electoral:

- Voter Eligibility credentials issued to all Phase 1 enrolled citizens
- Diaspora enrollment portal and embassy network operational
- Remote voting capability deployed and penetration tested
- International observer access API deployed
- USSD voter eligibility check live for rural low-connectivity access
- Independent audit of electoral module — minimum 2 international firms — complete minimum 14 days before referendum

Milestones — Justice:

- [Day 60] Anonymous testimony system operational and tested
- [Day 60] Conditional amnesty workflow operational
- Judicial escrow key ceremony completed with relevant judicial authorities

Scale: 10 million enrollments by Month 4 (full urban coverage).

29.4 Phase 3 — National Rollout (Months 4–12)

Milestones:

- Mobile enrollment units deployed to all 31 provinces (minimum 50 units; 200 recommended)
- All 11 credential types operational
- Pension and subsidy payment integration complete
- Healthcare and pharmacy verification live
- Full minority language support: Kurdish (Sorani + Kurmanji), Azerbaijani, Arabic, Baluchi
- Private sector verifier program launched: banks, telecoms, insurance
- Physical card production and distribution nationwide

Scale: 50 million enrollments by Month 12.

29.5 Phase 4 — Full Coverage (Months 12–24)

Milestones:

- Diaspora enrollment complete through full embassy network in 50+ countries
- Physical card distribution complete for non-smartphone population
- International interoperability framework with partner countries
- Post-quantum migration: all long-term credentials migrated to Dilithium3
- System optimization based on 12 months of production telemetry
- Full open-source publication of all components with public audit

Scale: 85+ million enrollments by Month 24 (full coverage; residual cases through ongoing mobile unit operations).

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30. Governance Structure

30.1 National Identity Authority (NIA)

The NIA is the governing body for INDIS. Its mandate:

- Issue and revoke credentials
- Manage the root cryptographic keys under multi-party control
- Operate the core identity services
- Publish the public ZK verification keys and audit logs
- Enforce the data minimization and privacy requirements
- Report to the Parliamentary Oversight Committee

Critical structural requirement: The NIA must be established by legislation, with its mandate and constraints legally defined. Future governments should not be able to expand the NIA's data access powers by administrative decision alone — any change to the privacy guarantees must require new legislation.

30.2 Parliamentary Oversight Committee

A Parliamentary Oversight Committee with technical staff receives:

- Monthly aggregate statistics (enrollment, verification, revocation volumes)
- Automatic alerts on any Level 4 emergency override
- Quarterly security audit reports
- Immediate notification of any data breach or security incident

The Committee has read-only access to sanitized audit logs. It has no administrative access to the system and cannot access individual citizen data.

30.3 Independent Ombudsman

An independent Identity Ombudsman handles:

- Citizen complaints about data access, correction requests, and privacy violations
- Appeals from citizens whose enrollment was rejected
- Complaints about unauthorized verifier behavior
- Annual public reporting on complaint volume and resolution

30.4 International Audit Partners

The following international organizations are recommended as audit partners:

- **Estonia e-Governance Academy** — technical audit; Estonia has 25 years of experience building exactly this type of sovereign digital identity infrastructure
- **UNDP Digital Finance** — program governance and inclusivity audit
- **ODIHR (OSCE Office for Democratic Institutions)** — electoral module audit and referendum observation
- **FATF-compliant national FIU** — AML/KYC compliance review for private sector verifier program

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31. International Standards Compliance

INDIS implements the following international standards, which are the basis for international recognition and interoperability:

Standard	Scope	Why It Matters
W3C DID Core 1.0	Decentralized Identifiers	Universal format recognized by all major identity systems
W3C VC Data Model 2.0	Verifiable Credentials	International credential format; required for cross-border recognition
OpenID Connect 4 VP	Credential presentation	Required for private sector verifier integration
ISO/IEC 18013-5	Mobile digital identity	International mobile ID standard; recognized at borders
ICAO 9303	Physical travel document	Required for physical card recognition at all international borders
ISO/IEC 30107-3	Biometric liveness detection	Required for international biometric quality recognition
ISO/IEC 29794	Biometric sample quality	International biometric standard
FIPS 140-2 Level 3	Hardware security module	International HSM standard; required for financial sector interoperability
NIST PQC (FIPS 203/204/205)	Post-quantum cryptography	Future-proof security standard; increasingly required by partner governments

Standard	Scope	Why It Matters
WCAG 2.1 Level AA	Accessibility	International web accessibility standard

32. Policy Decisions Required

The following decisions require explicit input from transitional government leadership. They are listed in order of timeline impact — earlier items block later deployments.

#	Decision	Options	Deadline	Recommendation
1	**Diaspora referendum participation**	Full participation / Mehestan elections only / Not eligible	Before Phase 2 launch	Full participation; consistent with Emergency Phase Booklet
2	**Social attestation voting eligibility**	Eligible immediately (with review) / Eligible after upgrade / Not eligible for referendum	Before Phase 2	Eligible after secondary review; defined appeal process
3	**Physical card first-issuance fee**	Free for all / Means-tested free / Fee for all	Before Phase 3	Free for first issuance; government commitment to universal access
4	**International audit partner selection**	Estonia / UNDP / UN OCHA / others	Before Phase 0 ends	Estonia + UNDP + ODIHR for electoral
5	**ZK trusted setup ceremony participants**	Iranian participants only / International multi-party	Before Phase 0 ends	International multi-party; include diaspora; maximum legitimacy
6	**Private sector verifier fee model**	Annual license (graduated by size) / Per-verification fee / Free	Before Phase 3	Annual license; creates sustainable revenue for system maintenance
7	**Data retention period after death**	10 years / 25 years / 50 years	Before Phase 1	25 years; balances justice needs with storage costs
8	**Biometric SDK: open-source vs. commercial**	Open-source (ArcFace + BOZORTH3) / Commercial with audit rights	Before Phase 1	Open-source first; commercial only if ISO accuracy targets unmet in testing

#	Decision	Options	Deadline	Recommendation
9	**Minor enrollment age cutoff**	15 / 16 / 18	Before Phase 1	16, with parental consent; aligns with education and healthcare needs
10	**Credential delegation scope**	Full delegation / Credential-type specific / No delegation	Before Phase 3	Credential-type specific; healthcare and pension delegation only

33. Risk Register

Risk	Probability	Impact	Mitigation	Owner
Biometric deduplication accuracy at 88M scale	Medium	High	Testing at full scale in isolated environment before Phase 3; accuracy targets contractually binding	NIA technical
Electoral module software bug affecting referendum	Low	Critical	Independent audit 14 days before referendum; end-to-end verifiability test publicly demonstrated	Electoral Commission + auditors
Regime remnant infiltration of enrollment agent network	Medium	High	Agent credential binding; anomaly detection; supervisory review for social attestation	NIA security
Nation-state cyberattack on electoral infrastructure	Medium	Critical	Air-gapped backups; geographically distributed infrastructure; international security partnerships	NIA security + international partners
Diaspora enrollment capacity at embassies	High	Medium	Prioritize embassies in largest diaspora countries; diaspora portal for pre-enrollment and scheduling	MFA + NIA
Minority language quality causing exclusion	Medium	High	Native speaker review for all UI text; community testing before Phase 3 launch	NIA + community liaisons
	Medium	Medium		NIA + MCIT

Risk	Probability	Impact	Mitigation	Owner
Feature phone USSD interoperability with all operators			Early telecom engagement; USSD specification standardized across operators	
Physical card supply chain	Medium	Medium	Identify card manufacturer with audit rights before Phase 3; no foreign-controlled card production	MFA + NIA
Post-quantum timeline compression	Low	High	Dilithium3 already implemented and tested; `pqc-migrate` tool ready; migration can begin at 30 days notice	NIA technical
Public trust deficit	High	High	Open-source code; public security audits; parliamentary oversight; privacy control center prominently accessible	NIA + parliamentary committee

34. Cost and Sustainability Model

34.1 Capital Costs

Component	Phase	Estimate Range	Notes
Server infrastructure (3 data centers)	Phase 0	$$–$$$	Sovereign hosting; no cloud costs
HSM cluster (FIPS 140-2 L3)	Phase 0	$$	3-5 HSM units for redundancy
Mobile enrollment units	Phase 1–3	$$–$$$	50–200 units; vehicles + hardware
Enrollment center hardware (biometric)	Phase 1	$	50+ centers × biometric station
Security audit (initial)	Phase 0	$	2 international firms
ZK trusted setup ceremony	Phase 0	$	Logistics; international observers

Component	Phase	Estimate Range	Notes
Embassy enrollment hardware	Phase 2	$$	50+ embassies × enrollment kit

34.2 Operational Costs

Item	Annual Estimate	Notes
Infrastructure operations	$$$	Power, hardware maintenance, staff
Security audits (ongoing)	$	Quarterly pen testing; annual full audit
Bug bounty program	$	Based on severity-weighted payouts
Mobile unit operations	$$	Fuel, maintenance, agent salaries
International observer access	$	API infrastructure; observer support

34.3 Revenue Streams for Sustainability

Source	Model	Notes
Private sector verifier licensing	Annual fee, graduated by organization size	Banks, insurance, telecoms
Card replacement fees	Per card	First issuance free; replacement charged
API access for high-volume commercial verifiers	Tiered by volume	Above free-tier threshold
International interoperability fees	Per-country bilateral agreement	Phase 4

The system must not be dependent on verification fees for core operational funding. Core funding should come from government appropriation. Revenue streams supplement but do not substitute for sovereign funding commitment.

APPENDIX A: FUNCTIONAL REQUIREMENTS REFERENCE

Full functional requirements are organized by the FR-XXX numbering system inherited from v1.0, with updates to reflect the completed implementation.

FR-001: Enrollment Processing

ID	Requirement	Priority	Implementation Status
FR-001.1	System SHALL process enrollment via three pathways: Standard, Enhanced, Social Attestation	MUST	[DONE] Complete
FR-001.2	Biometric deduplication SHALL complete within 90 seconds under normal load	MUST	[DONE] Complete
FR-001.3	System SHALL generate a DID conforming to W3C DID Core 1.0 for each enrolled individual	MUST	[DONE] Complete
FR-001.4	Private keys SHALL be generated on citizen's device only; government servers SHALL NEVER hold citizen private keys	MUST	[DONE] Complete
FR-001.5	System SHALL support minor enrollment through parent/guardian with linked guardian credential	MUST	[DONE] Complete
FR-001.6	System SHALL issue a temporary enrollment receipt credential immediately after biometric capture	MUST	[DONE] Complete
FR-001.7	System SHALL support bulk enrollment for institutional populations	SHOULD	[DONE] Complete
FR-001.8	Social attestation enrollment SHALL require minimum 3 co-attestors with valid Tier 2+ credential	MUST	[DONE] Complete
FR-001.9	Diaspora enrollment SHALL be available through embassy network with full credential parity	MUST	[DONE] Complete

FR-002: Credential Types and Lifecycle

ID	Requirement	Priority	Status
FR-002.1	11 credential types as specified in Section 10.2	MUST	[DONE] Complete
FR-002.R1	Revocation propagation to all verifier nodes: ≤ 60 seconds	MUST	[DONE] Complete
FR-002.R2		MUST	

ID	Requirement	Priority	Status
	Revocation registry SHALL be on-chain, checkable without querying central identity DB		[DONE] Complete
FR-002.R3	Selective disclosure MUST be supported for all credential types	MUST	[DONE] Complete
FR-002.R4	Expiry notifications: 30 days, 7 days, 1 day before expiry	SHOULD	[DONE] Complete

FR-003: Zero-Knowledge Proof System

ID	Requirement	Priority	Status
FR-003.1	ZK proof generation (standard): target 2s, maximum 5s on mid-range 2022 Android	MUST	[DONE] Complete
FR-003.2	ZK proof generation (electoral STARK): target 5s, maximum 15s	MUST	[DONE] Complete
FR-003.3	Proof verification at terminal: target 200ms, maximum 500ms	MUST	[DONE] Complete
FR-003.4	ALL ZK circuit code SHALL be open-source and publicly audited before production	MUST	Audit pending
FR-003.5	Formal verification of ZK circuits before production deployment	MUST	Pending

FR-004: Biometric Management

ID	Requirement	Priority	Status
FR-004.1	Face recognition + liveness detection required for all enrollments	MUST	[DONE] Complete (dev model)
FR-004.2	Fingerprint required; minimum 4 fingers; fallback for occupational wear	MUST	[DONE] Complete (dev model)
FR-004.3	False Match Rate ≤ 0.0001% (ISO/IEC 29794)	MUST	Production model pending
FR-004.4	Biometric templates encrypted AES-256-GCM; HSM-managed keys	MUST	[DONE] Complete
FR-004.5	Biometric data NEVER shared with any foreign entity	MUST	

ID	Requirement	Priority	Status
			[DONE] Architectural constraint

FR-005 through FR-017 requirements from the v1.0 PRD remain in force as stated, with implementation status as follows:

FR Range	Area	Status
FR-005 to FR-008	Citizen app platform, language, card display, privacy center	[DONE] Complete
FR-009 to FR-011	Government portal, electoral module, justice module	[DONE] Complete (portal ~98%)
FR-012 to FR-013	Verifier registration, result display	[DONE] Complete
FR-014	Audit service	[DONE] Complete
FR-015	USSD/SMS fallback	[DONE] Complete (telecom integration pending)
FR-016	Physical card	[DONE] Complete (NFC APDU encoding pending)
FR-017	Accessibility	[DONE] Complete (WCAG audit pending)

APPENDIX B: GLOSSARY

Term	Definition
DID	Decentralized Identifier — a W3C standard for a globally unique identifier that is under the control of the subject (the citizen), not a registry authority
Verifiable Credential (VC)	A cryptographically signed digital credential conforming to W3C VC Data Model 2.0 — the digital equivalent of a government-issued document
ZK-SNARK	Zero-Knowledge Succinct Non-interactive Argument of Knowledge — a mathematical proof system that proves the truth of a statement without revealing why it is true

Term	Definition
ZK-STARK	Zero-Knowledge Scalable Transparent Argument of Knowledge — a post-quantum-secure proof system requiring no trusted setup ceremony
Groth16	The most widely deployed SNARK proof system; produces the smallest proofs and verifies fastest; requires a trusted setup ceremony
Bulletproofs	A proof system with no trusted setup, optimal for range proofs and anonymous testimony
HSM	Hardware Security Module — a tamper-resistant physical device that stores cryptographic keys and performs signing operations; keys cannot be extracted
CRYSTALS-Dilithium3	The NIST-selected post-quantum digital signature algorithm; secure against quantum computer attacks
INDIS	Iran National Digital Identity System
NIA	National Identity Authority — the governing body responsible for operating INDIS
mTLS	Mutual TLS — both sides of a connection present certificates and authenticate each other; prevents impersonation of services
Social Attestation	The enrollment pathway for undocumented citizens: three enrolled community members cryptographically co-sign an identity attestation
Selective Disclosure	Sharing only specific credential attributes needed for a transaction, rather than the full credential
Nullifier	An anonymous, unlinkable cryptographic marker proving that a vote was cast, without revealing who cast it or how
Receipt-Freeness	A property of voting systems that prevents a voter from proving to a third party how they voted, eliminating coercion
Differential Privacy	A mathematical technique for publishing aggregate statistics in a way that provably protects individual privacy
OpenAPI 3.0	An internationally standard machine-readable specification for REST APIs; enables automatic client code generation
gRPC	A high-performance remote procedure call framework; all internal INDIS service communication uses gRPC
Hyperledger Fabric	A permissioned enterprise blockchain platform maintained by the Linux Foundation; no foreign validator control
Kart-e Melli	Iranian national identity card

Term	Definition
Shenasnameh	Iranian civil registration birth certificate
Solar Hijri	The Iranian solar calendar (Shamsi); the primary date system in INDIS
Vazirmatn	An open-source Persian typeface optimized for screen rendering, maintained by the Iranian open-source community

APPENDIX C: STANDARDS AND REFERENCES

Standard	Full Name	Scope in INDIS
W3C DID Core 1.0	Decentralized Identifiers	DID generation, resolution, and anchoring
W3C VC Data Model 2.0	Verifiable Credentials Data Model	All 11 credential types
OpenID4VP	OpenID Connect for Verifiable Presentations	Private sector verifier integration (Phase 3)
ISO/IEC 18013-5	Mobile Driving Licence Application	Digital identity interoperability at borders
ICAO 9303	Machine Readable Travel Documents	Physical identity card specification
ISO/IEC 30107-3	Biometric Presentation Attack Detection	Liveness detection for enrollment
ISO/IEC 29794	Biometric Sample Quality	Biometric deduplication accuracy targets
FIPS 140-2 Level 3	Security Requirements for Cryptographic Modules	HSM specification
NIST FIPS 204	Module-Lattice-Based Digital Signature Standard (Dilithium)	Post-quantum signature
NIST FIPS 203	Module-Lattice-Based Key Encapsulation (Kyber)	Post-quantum key exchange (Phase 4)
WCAG 2.1 Level AA	Web Content Accessibility Guidelines	Citizen app and portal accessibility

Standard	Full Name	Scope in INDIS
RFC 8032	Edwards-Curve Digital Signature Algorithm (Ed25519)	Current credential signatures
RFC 8446	TLS 1.3	All transport security
Circom 2.0	ZK Circuit Language	Age, citizenship, and voter eligibility circuits

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APPENDIX D: CURRENT BUILD STATUS

The INDIS system is substantially complete. As of March 2026, the implementation status is:

Component	Completion	Remaining
Shared Go libraries (12 packages)	100%	None
Backend Go services (15 services)	97–99%	HSM gateway JWT wiring; govportal bulk operation
ZK proof engine (Rust)	92%	Formal verification of circuits; production trusted setup
AI biometric service (Python)	60%	Production CNN model (FaceNet/ArcFace); NIST fingerprint SDK
Blockchain chaincode (Go)	95%	Code complete; Fabric network deployment pending
API specifications (OpenAPI + Proto)	100%	None
Infrastructure (Helm, Terraform, CI/CD)	97%	Production HSM integration; Fabric network
Citizen PWA	95%	E2E test coverage ≥50%
Government portal	98%	Role PUT handler; bulk operation execution
Verifier terminal PWA	90%	E2E tests
Android app	95%	Detox E2E tests
iOS app	90%	Xcode project; Rust ZK bridge

Component	Completion	Remaining
HarmonyOS app	95%	Hardware device testing
Diaspora portal	95%	Electoral eligibility rules (policy TBD)
System-wide	**~97%**	Production infrastructure, biometric AI, ZK ceremony, security audit

What remains before production deployment:

1. **Security audits** — 2 independent international firms; penetration testing of all surfaces; formal ZK circuit verification
2. **Production infrastructure** — Hyperledger Fabric network (21+ nodes); HashiCorp Vault HSM cluster; 3-datacenter deployment
3. **ZK trusted setup ceremony** — Multi-party, international observers
4. **Production biometric AI** — FaceNet/ArcFace ONNX; NIST fingerprint SDK; ISO accuracy certification
5. **Telecom integration** — USSD short code with national operators
6. **Notification providers** — SMS, push, email delivery for credential lifecycle notifications

All of these are infrastructure and procurement decisions — the software is ready to be deployed against them. The system is not waiting for additional development; it is waiting for the institutional and infrastructure decisions that only the transitional government can make.

Final Note to Policy Decision-Makers

This document describes a system built for a specific historical moment: the transition of Iran from an authoritarian state to a democratic one. Every design decision in INDIS was made with that context in mind.

The system is privacy-first because Iranians have lived under surveillance for 45 years and they deserve a government that treats their information as belonging to them, not to the state.

The system is inclusive because millions of Iranians were systematically excluded from official existence by the previous government, and any legitimate new government must begin by recognizing that every Iranian has a right to an identity.

The system is open-source and internationally auditable because trust must be earned through transparency, not demanded through authority.

The system is built on cryptographic proofs rather than institutional promises because cryptographic proofs hold regardless of who runs the government next.

INDIS is not just an identity system. It is a statement about the kind of country Iran intends to become.

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Source code for all cryptographic components is open-source and publicly auditable. The code is the proof.